

Household Services Application Report

Author Details

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Background: My name is Aneeqha Saba. I am 20 years old, from Tamil Nadu. I graduated from the Indian School Al Wadi Al Kabir in Oman and am currently in my 3rd semester of a diploma in Data Science and Application at IIT Madras.

Project Details

The Household Services Application is a platform designed to connect customers with professionals for various home-related services. The application includes 3 roles, Admin, Service Professionals, and Customers, each with distinct features and access levels. The main functionalities cover service requests, professional profiles, admin management tools, and customer interactions.

Problem Statement and Approach

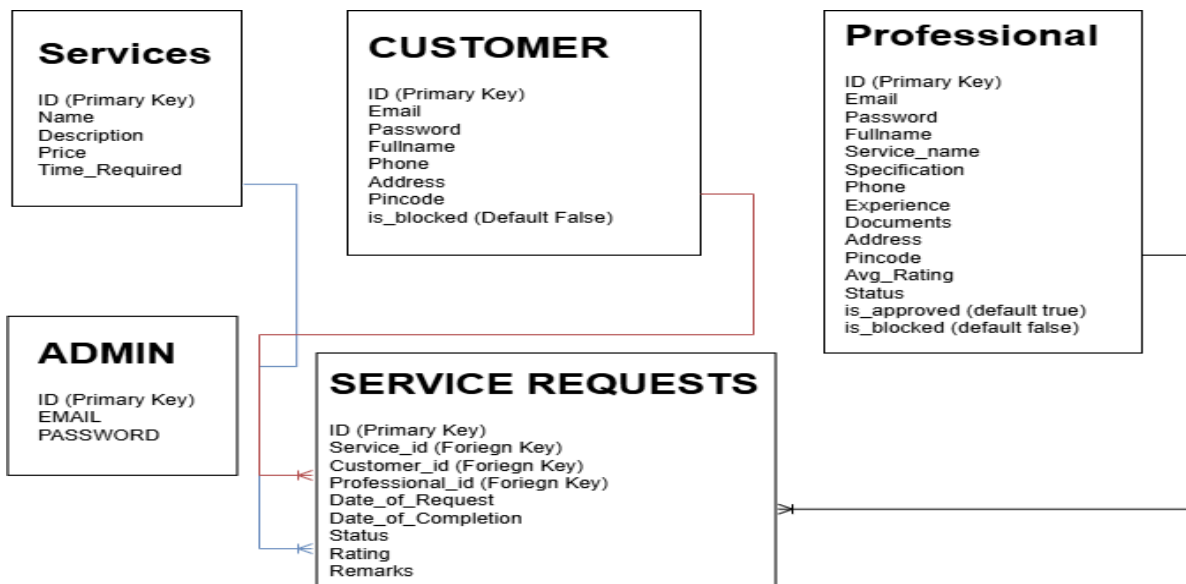
Problem Statement requires us to develop a web-based application that facilitates the connection between customers and service professionals for household tasks, while providing a comprehensive admin interface for managing users and services.

Approach: The solution was developed using a modular approach, following the Model-View-Controller (MVC) design pattern for clear structure and maintainability. The backend was built with Flask to handle routing, HTTP requests, and server-side logic, and SQLAlchemy was used for database interactions.

Frameworks and Libraries Used

- Flask: Micro web framework for web application development.
 - `render_template`: Dynamically renders HTML templates.
 - `request`, `redirect`, `url_for`: Manages HTTP requests, user redirections, and URL generation.
 - `session`: Stores user session data.
 - `flash`: Displays notifications for user actions.
 - `current_app`: Accesses the Flask application instance.
 - `send_from_directory`: Serves files for downloads.
- SQLAlchemy: Object Relational Mapper for managing database connections and queries.
- Werkzeug (secure_filename): Ensures safe file uploads.
- os (Operating System module): Interacts with file paths and directories.
- datetime, date (Python): Handles date/time data.
- Matplotlib: Generates visual charts for data representation.

DB Schema Design



API Design

The Household Services Application includes RESTful APIs for CRUD operations related to services. The endpoints allow:

- GET `/services/<id>`: Retrieve service details.
- POST `/services`: Create a new service entry.
- PUT `/services/<id>`: Update service details.
- DELETE `/services/<id>`: Delete a service entry.

Data validation is managed using `reqparse`, and unique service IDs are generated using a custom `generate_service_id()` function.

Architecture and Features

The application is structured using the MVC pattern, enabling a clean codebase and efficient management:

Models: Defined in `models.py`, representing database schema.

Controllers: Implemented within route modules to handle HTTP requests and connect models with templates.

Templates: Jinja2 templates in the `templates` directory for rendering dynamic HTML content.

Static Files: Stored in the `static` directory, including CSS for frontend styling.

Key Features:

Professional Approval: Only approved professionals can access their dashboards.

File Uploads: Uses Werkzeug for secure CV uploads by professionals.

Drive Link for Presentation Video - Presentation Video

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